

Data Visualisation

Untitled

Visualising time series data with `timetk`

Packages

```
library("timetk")
library("tidyquant")
library(tidyverse)
library(denguedatahub)
library(anomalize)
library(tsibble)
library(fable)
library(fabletools)
library(feasts) #STL decomposition
```

Data

```
data(srilanka_weekly_data)
colombo <- srilanka_weekly_data |>
  filter(district == "Colombo")
# create a proper date index column
colombo <- srilanka_weekly_data |>
  filter(district == "Colombo") |>
  mutate(
    start.date = mdy(start.date)
  )
```

Time Series Plot

```
colombo |>
  plot_time_series(
    .date_var = start.date,
    .value = cases,
    .interactive = FALSE
  ) +
  labs(
    title = "Weekly Dengue Cases in Colombo",
    x = "Year",
    y = "Cases"
  )
```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.

• Please use `linewidth` instead.

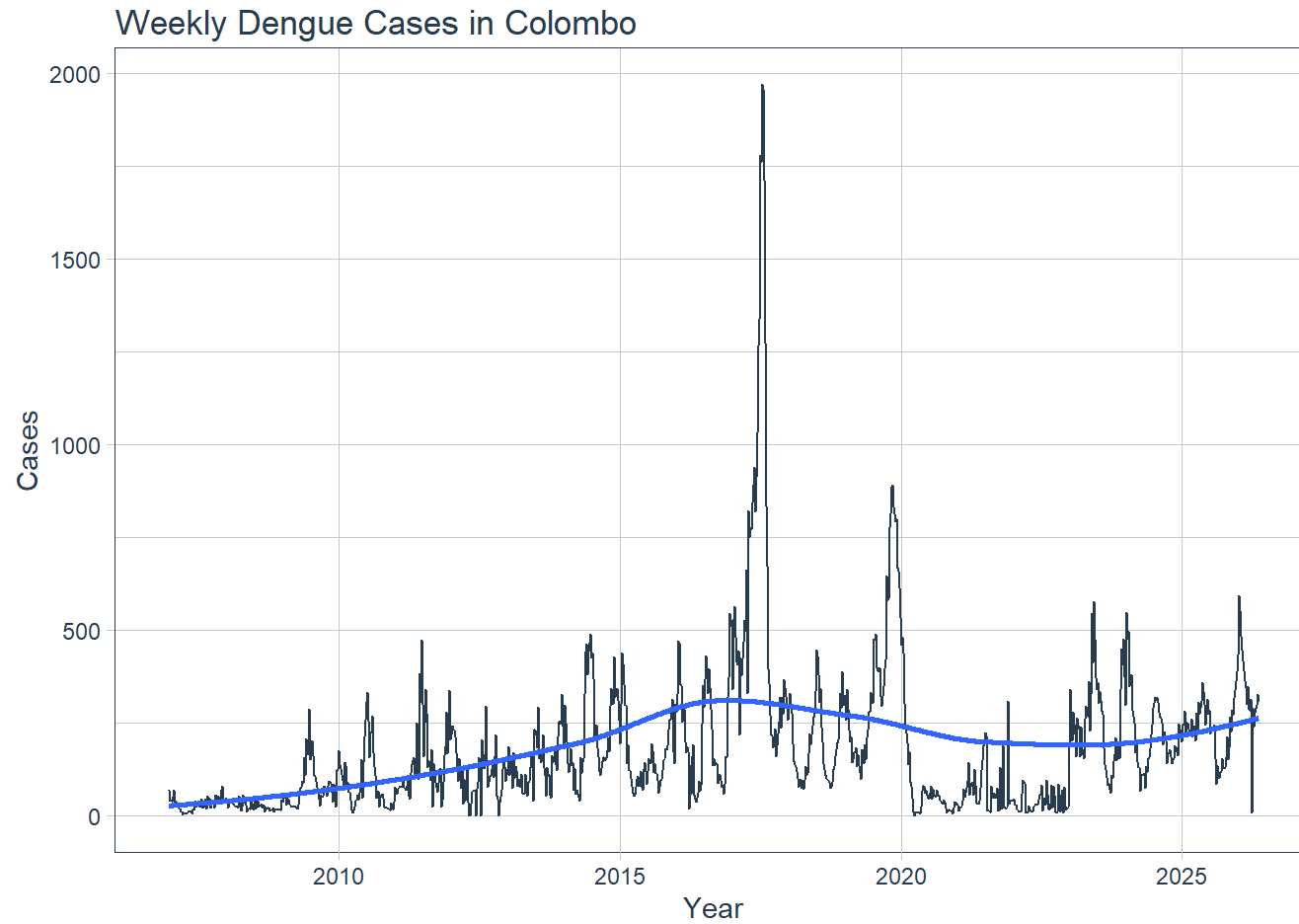
• The deprecated feature was likely used in the timetk package.

Please report the issue at

<<https://github.com/business-science/timetk/issues>>.

Ignoring unknown labels:

- colour : "Legend"

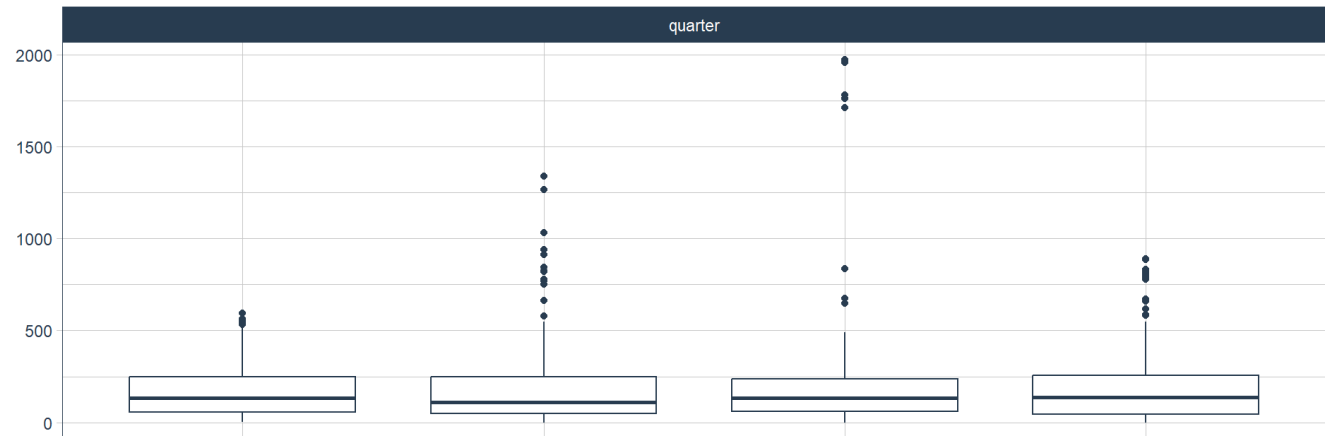
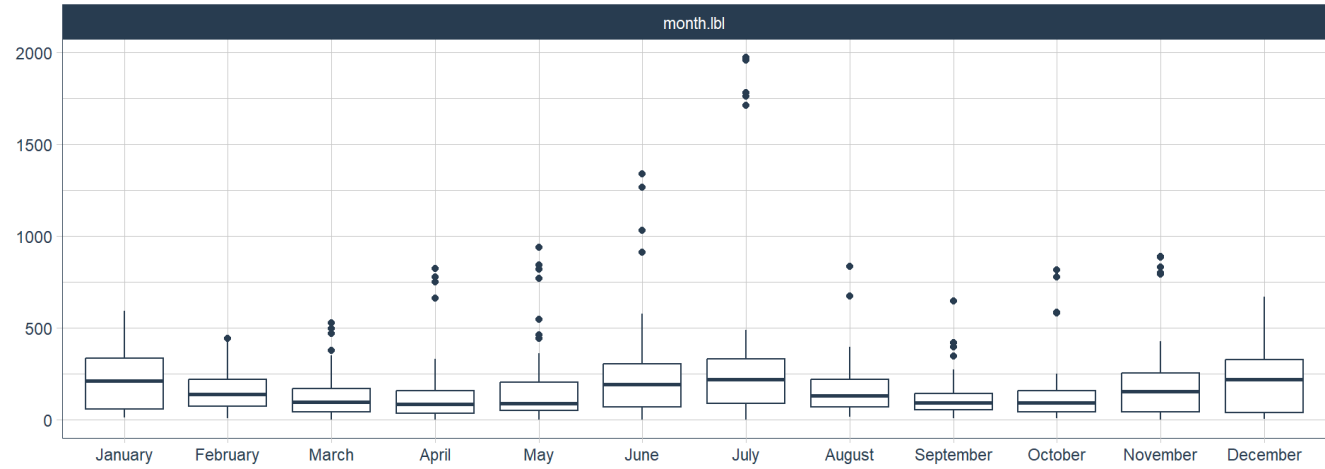
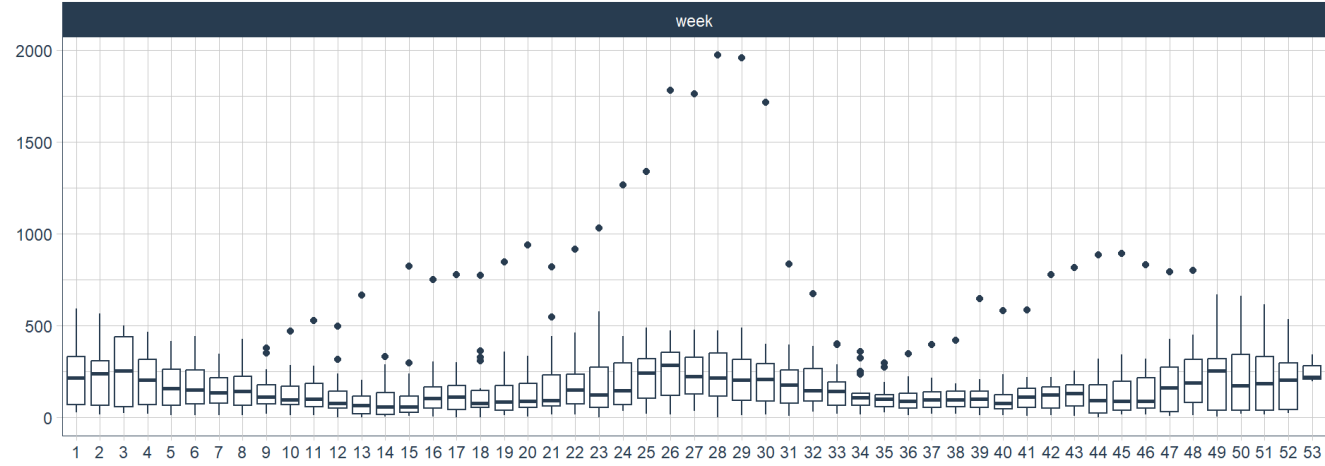


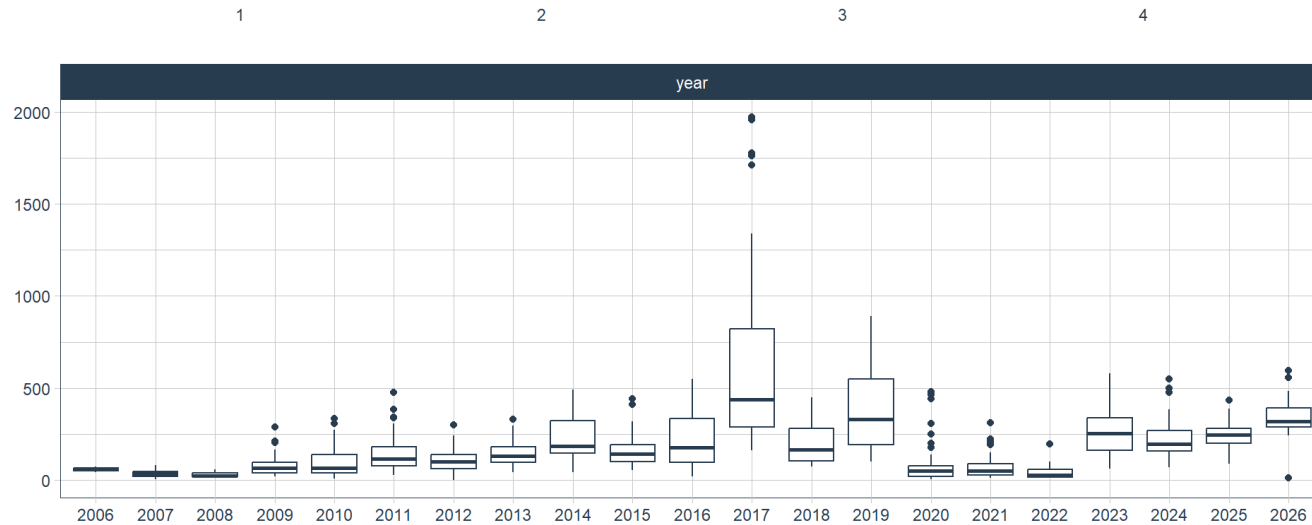
```
data(srilanka_weekly_data)
colombo <- srilanka_weekly_data |>
  filter(district == "Colombo")
# create a proper date index column
colombo <- srilanka_weekly_data |>
  filter(district == "Colombo") |>
  mutate(
    start.date = mdy(start.date)
  )
```

Seasonal plot

```
colombo |>  
  plot_seasonal_diagnostics(  
    .date_var = start.date,  
    .value = cases,  
    .interactive = FALSE  
  )
```

Seasonal Diagnostics





STL decomposition

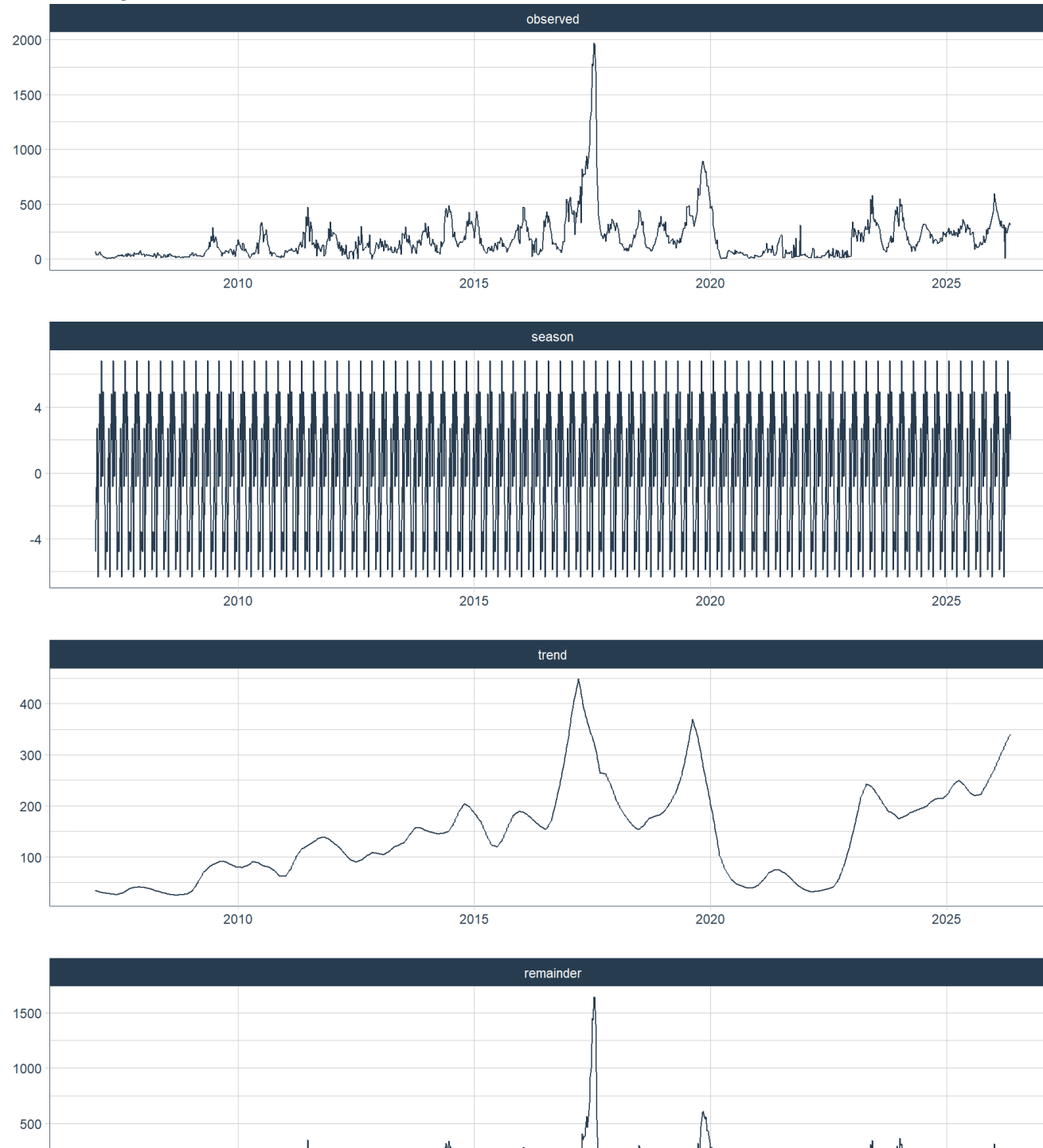
What do you think about the plot? Is this correct?

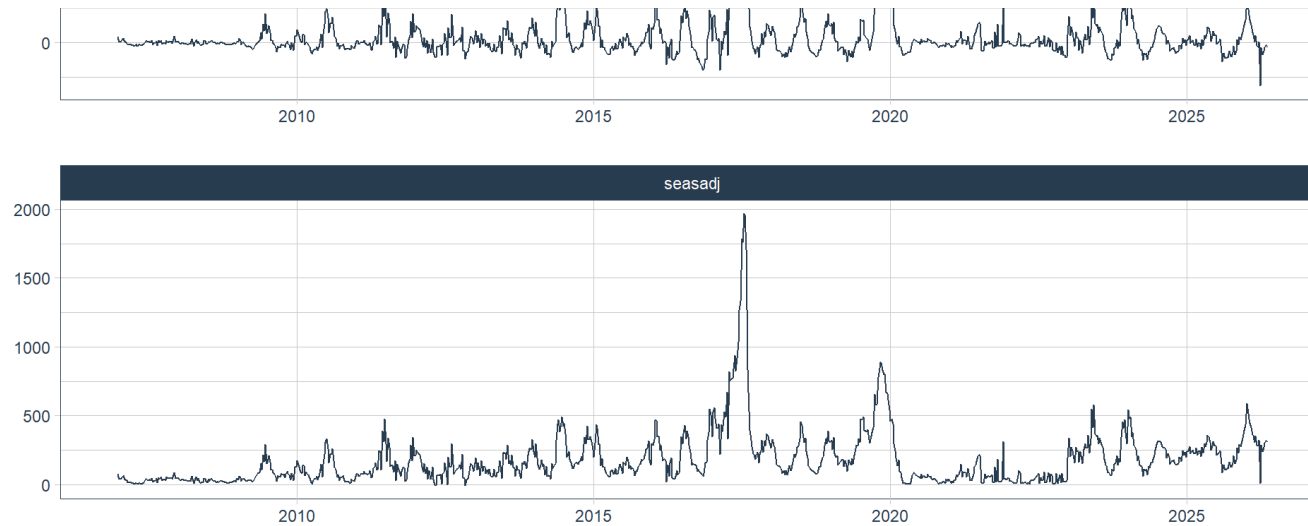
```
colombo |>  
  plot_stl_diagnostics(  
    .date_var = start.date,  
    .value = cases,  
    .interactive = FALSE  
  )
```

frequency = 13 observations per 1 quarter

trend = 53 observations per 1 year

STL Diagnostics





Let's fix the issue using tsibble and fabletools

What is wrong with the following code?

```
colombo_tsibble <- srilanka_weekly_data |>
  filter(district == "Colombo") |>
  mutate(
    start.date = mdy(start.date)
  ) |>
  arrange(start.date) |>
  as_tsibble(index = start.date)
colombo_tsibble
```

A tsibble: 991 x 6 [1D]

	year	week	start.date	end.date	district	cases
	<dbl>	<dbl>	<date>	<chr>	<chr>	<dbl>
1	2006	52	2006-12-23	12/29/2006	Colombo	71
2	2007	1	2006-12-30	1/5/2007	Colombo	40
3	2007	2	2007-01-06	1/12/2007	Colombo	43
4	2007	3	2007-01-13	1/19/2007	Colombo	38
5	2007	4	2007-01-20	1/26/2007	Colombo	52
6	2007	5	2007-01-27	2/2/2007	Colombo	69


```

7 2007      6 2007-02-03 2/9/2007  Colombo    36
8 2007      7 2007-02-10 2/16/2007 Colombo    42
9 2007      8 2007-02-17 2/23/2007 Colombo    28
10 2007     9 2007-02-24 3/2/2007  Colombo    20
# i 981 more rows

```

Let's fix the issue

```

colombo_tsibble <- srilanka_weekly_data |>
  filter(district == "Colombo") |>
  mutate(
    start.date = mdy(start.date),
    year_week = yearweek(start.date)
  ) |>
  arrange(year_week) |>
  as_tsibble(index = year_week)
colombo_tsibble

```

```

# A tsibble: 991 x 7 [1W]
   year week start.date end.date  district cases year_week
  <dbl> <dbl> <date>      <chr>      <chr>    <dbl>    <week>
1  2006   52 2006-12-23 12/29/2006 Colombo     71  2006 W51
2  2007    1 2006-12-30 1/5/2007   Colombo     40  2006 W52
3  2007    2 2007-01-06 1/12/2007 Colombo     43  2007 W01
4  2007    3 2007-01-13 1/19/2007 Colombo     38  2007 W02
5  2007    4 2007-01-20 1/26/2007 Colombo     52  2007 W03
6  2007    5 2007-01-27 2/2/2007   Colombo     69  2007 W04
7  2007    6 2007-02-03 2/9/2007   Colombo     36  2007 W05
8  2007    7 2007-02-10 2/16/2007 Colombo     42  2007 W06
9  2007    8 2007-02-17 2/23/2007 Colombo     28  2007 W07
10 2007    9 2007-02-24 3/2/2007   Colombo     20  2007 W08
# i 981 more rows

```

```
interval(colombo_tsibble)
```

```

<interval[1]>
[1] 1W

```

```
colombo_stl <- colombo_tsibble |>
  model(
    STL(cases ~ season(period = 52))
  )
colombo_stl
```

```
# A mable: 1 x 1
  `STL(cases ~ season(period = 52))`
      <model>
1      <STL>
```

```
colombo_components <- colombo_stl |>
  components()
colombo_components
```

```
# A dable: 991 x 7 [1W]
# Key:      .model [1]
# :      cases = trend + season_52 + remainder
```

	.model	year_week	cases	trend	season_52	remainder	season_adjust
	<chr>	<week>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	STL(cases ~ season(p...	2006 W51	71	22.6	24.7	23.7	46.3
2	STL(cases ~ season(p...	2006 W52	40	22.9	14.0	3.10	26.0
3	STL(cases ~ season(p...	2007 W01	43	23.1	24.4	-4.53	18.6
4	STL(cases ~ season(p...	2007 W02	38	23.4	19.7	-5.12	18.3
5	STL(cases ~ season(p...	2007 W03	52	23.7	20.4	7.92	31.6
6	STL(cases ~ season(p...	2007 W04	69	24.0	18.6	26.4	50.4
7	STL(cases ~ season(p...	2007 W05	36	24.2	4.55	7.21	31.5
8	STL(cases ~ season(p...	2007 W06	42	24.5	4.54	12.9	37.5
9	STL(cases ~ season(p...	2007 W07	28	24.8	1.73	1.47	26.3
10	STL(cases ~ season(p...	2007 W08	20	25.1	-18.4	13.3	38.4

```
# i 981 more rows
```

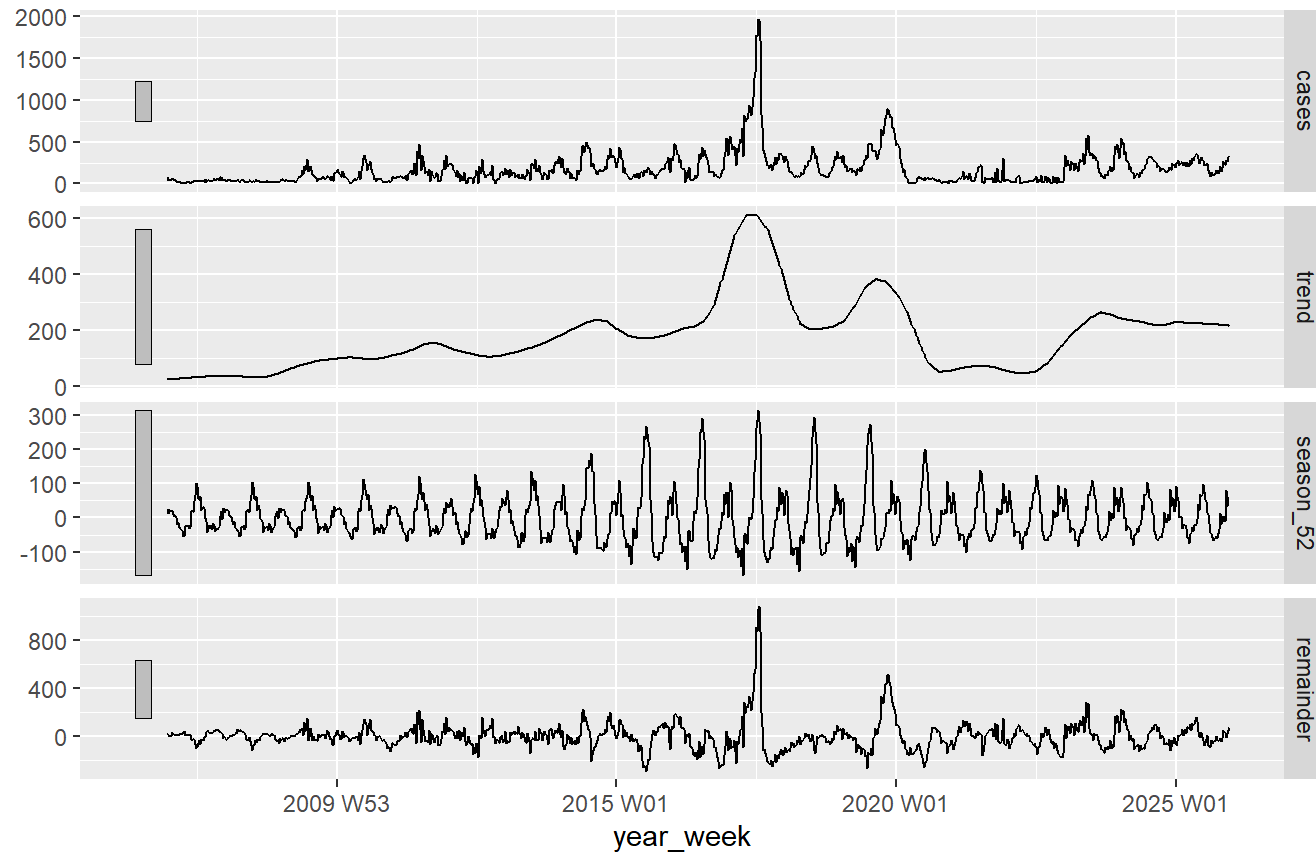
```
colombo_components |>
  autoplot()
```

Ignoring unknown labels:

- colour : ".model"
- fill : ".model"

STL decomposition

cases = trend + season_52 + remainder



Let's fix the issue with timetk

```
colombo %>%  
  plot_stl_diagnostics(  
    .date_var = start.date,  
    .value = cases,  
    .frequency = 52,  
    .trend = 104,
```

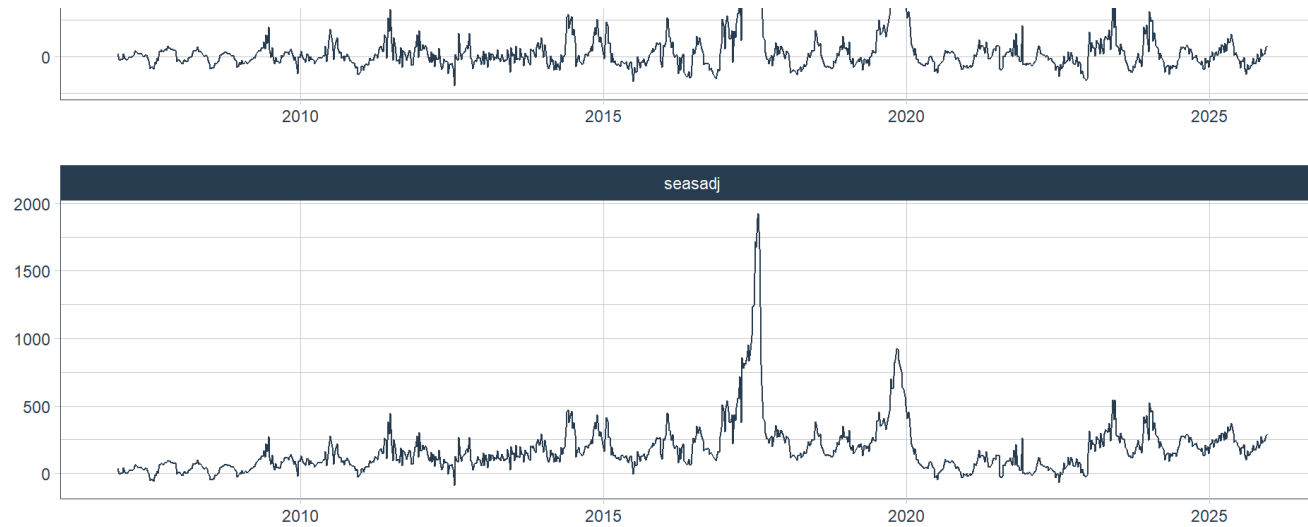
```
.interactive = FALSE  
)
```

frequency = 52 observations

trend = 104 observations

STL Diagnostics





Anomaly Detection

```
colombo %>%
  time_decompose(
    cases,
    method = "stl",
    frequency = 52
  ) %>%
  anomalise(
    remainder,
    method = "iqr"
  ) %>%
  time_recompose() %>%
  plot_anomalies(time_recomposed = TRUE)
```

Registered S3 method overwritten by 'tibbletime':

```
method from
seq.hms hms
```

Converting from tbl_df to tbl_time.

Auto-index message: index = start.date

```
frequency = 52 weeks
```

```
trend = 52 weeks
```

```
Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
```

```
i Please use tidy evaluation idioms with `aes()`.
```

```
i See also `vignette("ggplot2-in-packages")` for more information.
```

```
i The deprecated feature was likely used in the anomalize package.
```

```
Please report the issue at
```

```
<https://github.com/business-science/anomalize/issues>.
```

```
Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
```

```
i Please use the `linewidth` argument instead.
```

```
i The deprecated feature was likely used in the anomalize package.
```

```
Please report the issue at
```

```
<https://github.com/business-science/anomalize/issues>.
```

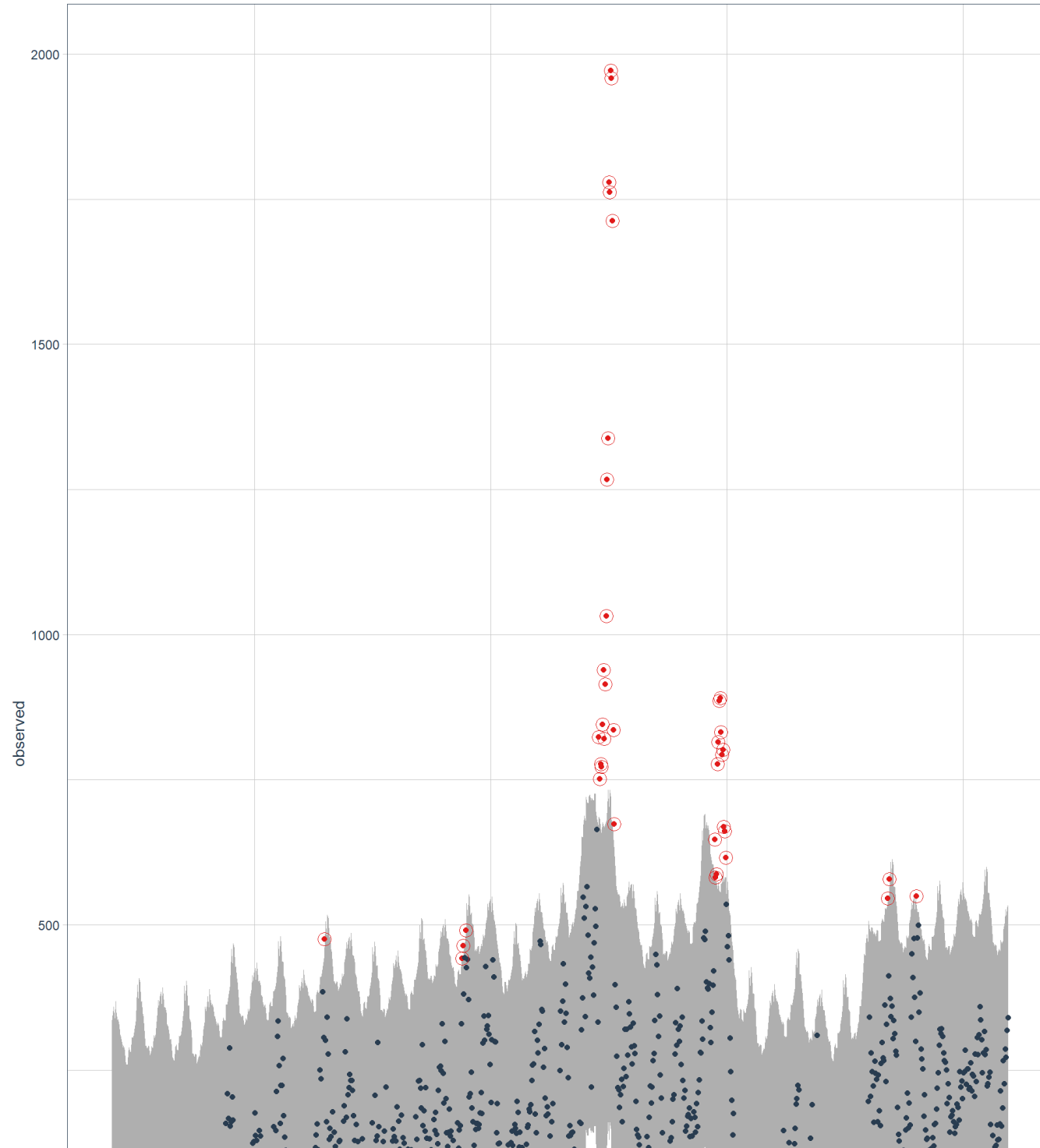
```
Warning: The `size` argument of `element_rect()` is deprecated as of ggplot2 3.4.0.
```

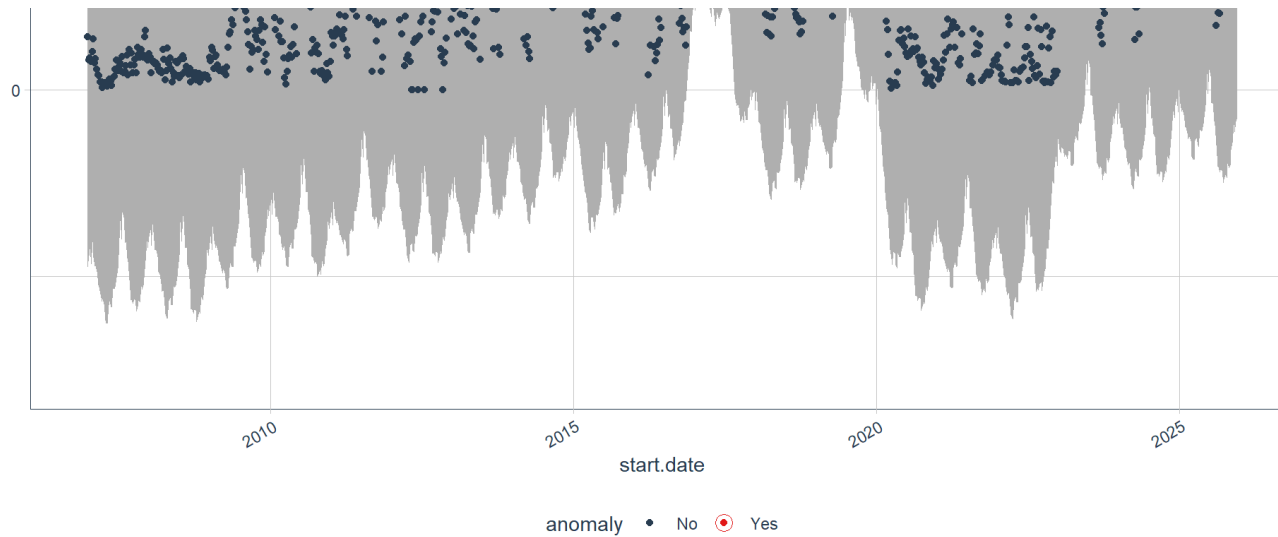
```
i Please use the `linewidth` argument instead.
```

```
i The deprecated feature was likely used in the anomalize package.
```

```
Please report the issue at
```

```
<https://github.com/business-science/anomalize/issues>.
```





Calendar Heatmap

```
colombo |>
  plot_time_series(
    .date_var = start.date,
    .value = cases,
    .color_var = year(start.date),
    .interactive = TRUE
  )
```

Warning: There were 5 warnings in `dplyr::mutate()`.

The first warning was:

! In argument: `.value\_smooth = auto\_smooth(...)`.

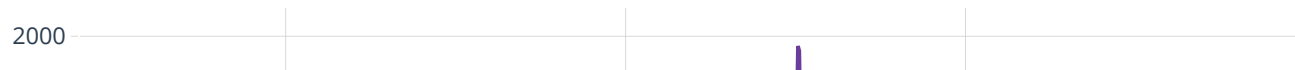
! In group 1: `.color\_mod = 2006`.

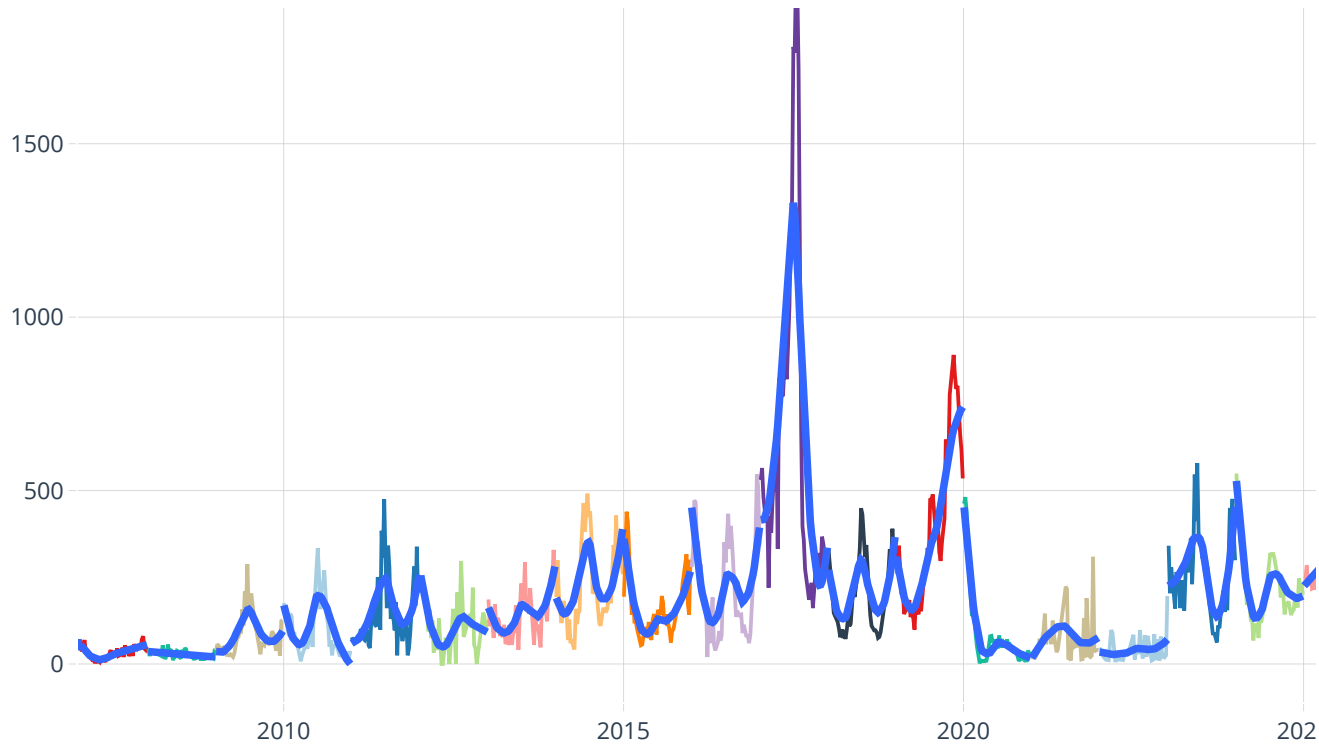
Caused by warning in `simpleLoess()`:

! span too small. fewer data values than degrees of freedom.

! Run `dplyr::last\_dplyr\_warnings()` to see the 4 remaining warnings.

Time Series Plot



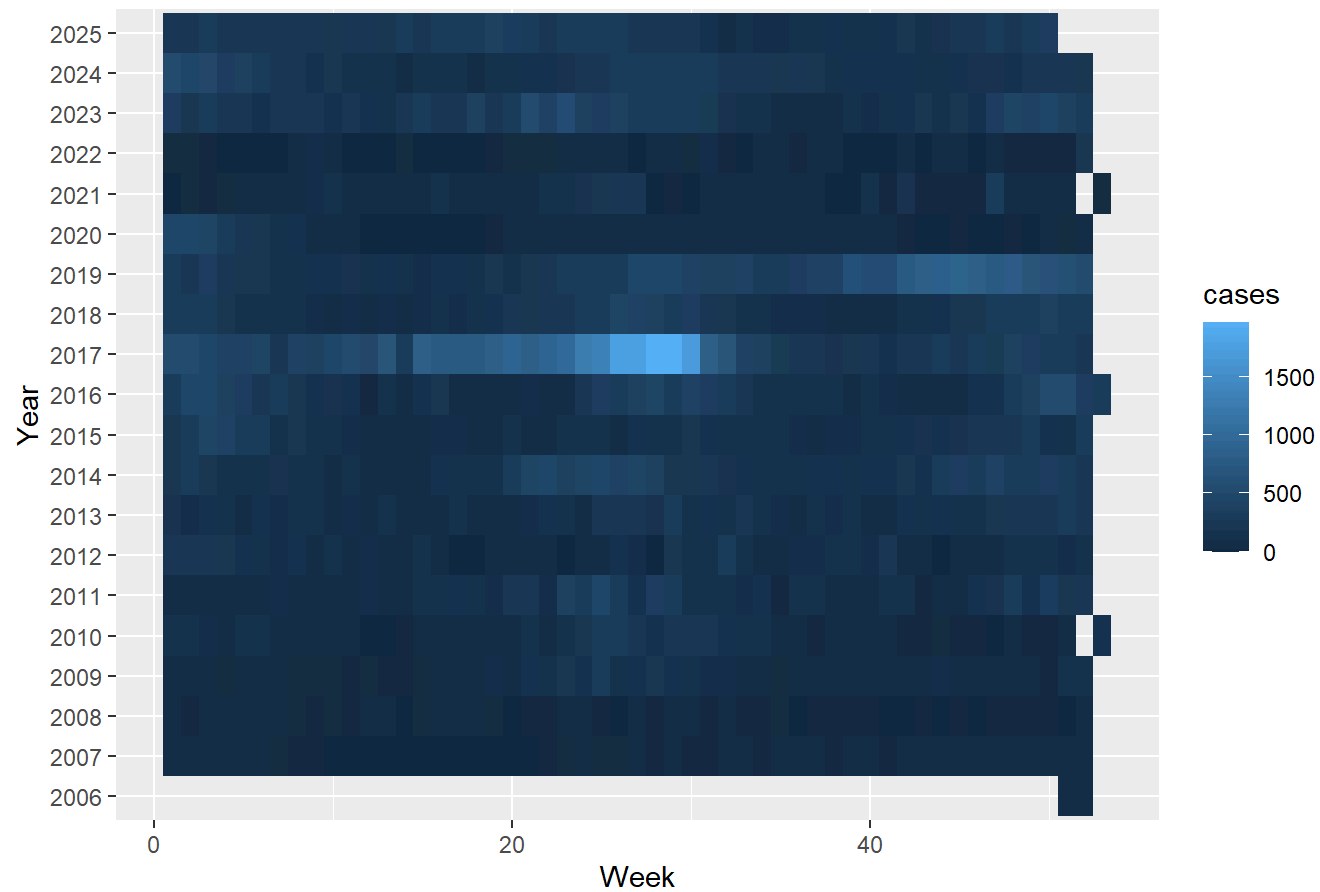


or

```
library(ggplot2)

colombo |>
  mutate(
    Year = year(start.date),
    Week = isoweek(start.date)
  ) |>
  ggplot(aes(Week, factor(Year), fill = cases)) +
  geom_tile() +
  labs(
    title = "Weekly Dengue Cases Heatmap",
    x = "Week",
    y = "Year"
  )
)
```

Weekly Dengue Cases Heatmap



Rolling Mean

```
library(dplyr)
library(slider)
library(timetk)

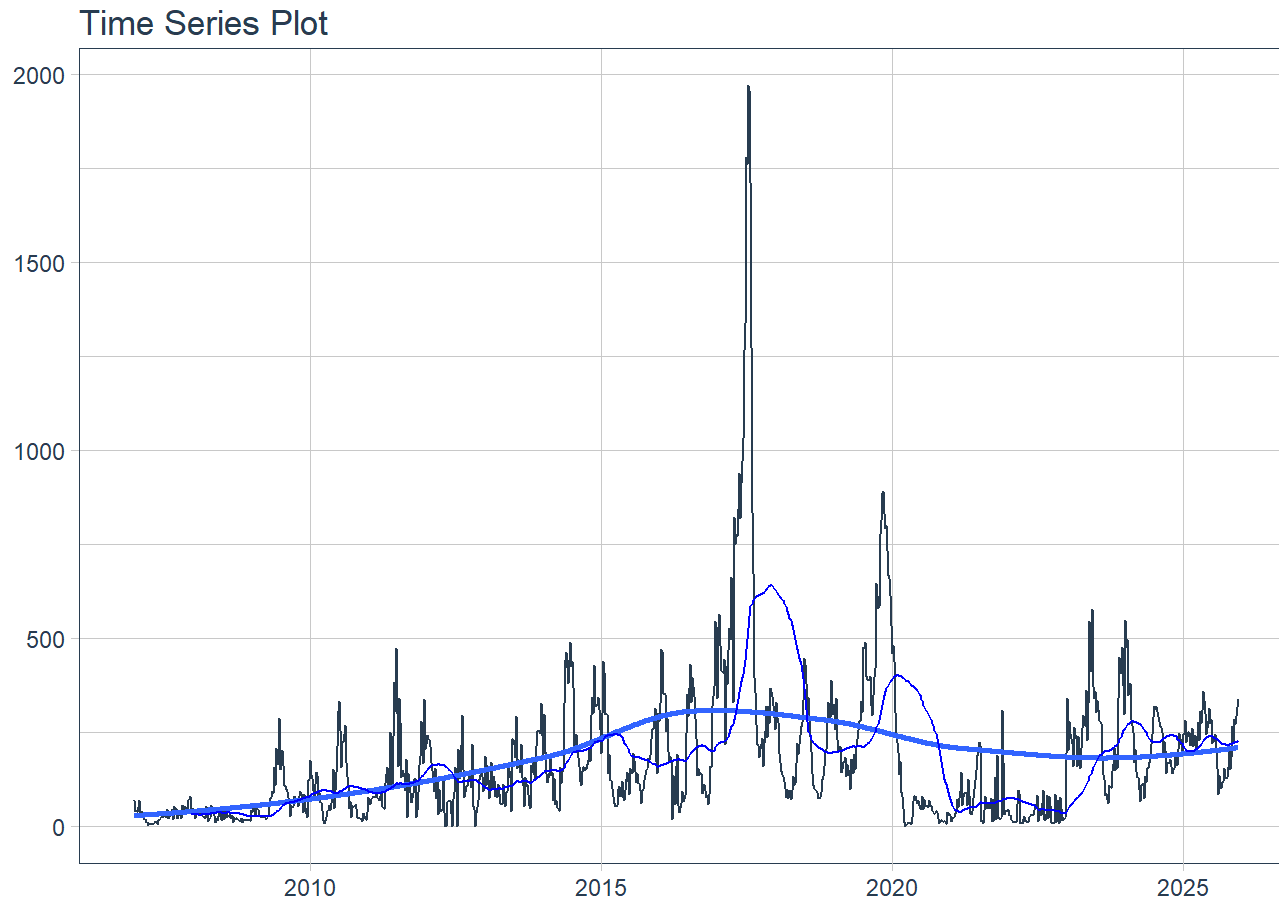
colombo |>
  mutate(
    MA52 = slide_dbl(cases,
                      mean,
```

```
      .before = 51,  
      .complete = TRUE)  
)  
|>  
plot_time_series(  
  .date_var = start.date,  
  .value = cases,  
  .interactive = FALSE  
) +  
geom_line(aes(y = MA52), color = "blue")
```

Ignoring unknown labels:

- colour : "Legend"

Warning: Removed 51 rows containing missing values or values outside the scale range (``geom_line()``).



Time Series Signature Exploration

```
colombo |>  
  tk_augment_timeseries_signature(start.date) |>  
  glimpse()
```

New names:

- `year` -> `year...1`
- `week` -> `week...2`

- `year` -> `year...9`
- `week` -> `week...29`

Rows: 991

Columns: 34

```
$ year...1 <dbl> 2006, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007,...
$ week...2 <dbl> 52, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, ...
$ start.date <date> 2006-12-23, 2006-12-30, 2007-01-06, 2007-01-13, 2007-01-20...
$ end.date <chr> "12/29/2006", "1/5/2007", "1/12/2007", "1/19/2007", "1/26/2...
$ district <chr> "Colombo", "Colombo", "Colombo", "Colombo", "Colombo", "Col...
$ cases <dbl> 71, 40, 43, 38, 52, 69, 36, 42, 28, 20, 19, 11, 11, 3, 10, ...
$ index.num <dbl> 1166832000, 1167436800, 1168041600, 1168646400, 1169251200,...
$ diff <dbl> NA, 604800, 604800, 604800, 604800, 604800, 604800, 604800,...
$ year...9 <int> 2006, 2006, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007,...
$ year.iso <int> 2006, 2006, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007,...
$ half <int> 2, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,...
$ quarter <int> 4, 4, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2,...
$ month <int> 12, 12, 1, 1, 1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, 4, 4, 4, 4, ...
$ month.xts <int> 11, 11, 0, 0, 0, 0, 1, 1, 1, 1, 2, 2, 2, 2, 2, 3, 3, 3, 3, ...
$ month.lbl <ord> December, December, January, January, January, January, Feb...
$ day <int> 23, 30, 6, 13, 20, 27, 3, 10, 17, 24, 3, 10, 17, 24, 31, 7,...
$ hour <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ minute <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ second <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ hour12 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ am.pm <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,...
$ wday <int> 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7,...
$ wday.xts <int> 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6,...
$ wday.lbl <ord> Saturday, Saturday, Saturday, Saturday, Saturday, Saturday,...
$ mday <int> 23, 30, 6, 13, 20, 27, 3, 10, 17, 24, 3, 10, 17, 24, 31, 7,...
$ qday <int> 84, 91, 6, 13, 20, 27, 34, 41, 48, 55, 62, 69, 76, 83, 90, ...
$ yday <int> 357, 364, 6, 13, 20, 27, 34, 41, 48, 55, 62, 69, 76, 83, 90...
$ mweek <int> 4, 5, 1, 2, 3, 4, 1, 2, 3, 4, 1, 2, 3, 4, 5, 1, 2, 3, 4, 1,...
$ week...29 <int> 51, 52, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, ...
$ week.iso <int> 51, 52, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, ...
$ week2 <int> 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0,...
$ week3 <int> 0, 1, 1, 2, 0, 1, 2, 0, 1, 2, 0, 1, 2, 0, 1, 2, 0, 1, 2, 0,...
```

```
$ week4      <int> 3, 0, 1, 2, 3, 0, 1, 2, 3, 0, 1, 2, 3, 0, 1, 2, 3, 0, 1, 2,...  
$ mday7      <int> 4, 5, 1, 2, 3, 4, 1, 2, 3, 4, 1, 2, 3, 4, 5, 2, 3, 4, 5, 1,...
```

Additional Reading

[See the plot](#)